

RHEL 9 Automated Patching with Email Notification

Prepared by: Rishabh Awasthi

🔧 Environment Details

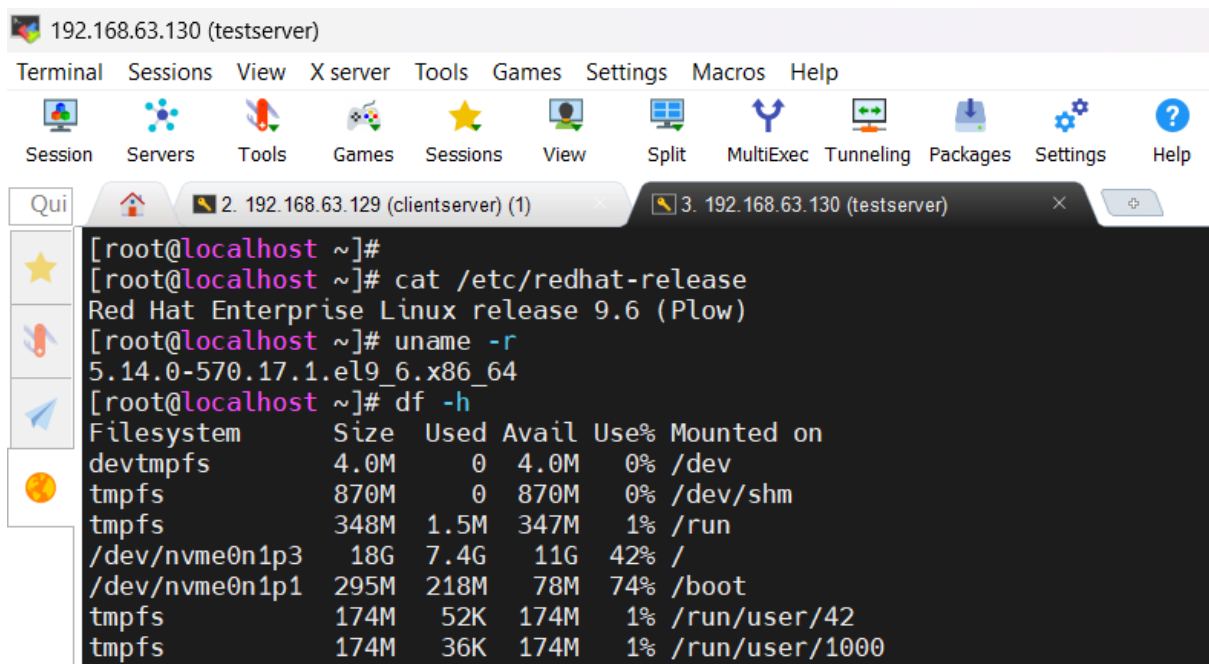
- **OS** : RHEL 9
- **Patch Tool** : DNF
- **Mail Utility** : Postfix + s-nail
- **Execution User** : root
- **Environment** : Non-Production

✅ Steps to Patch RHEL 9 Using an Automated Shell Script

🔒 Note: It is recommended to take a system backup or snapshot before applying patches in a production environment.

✅ Step 1: Verify OS Version, Kernel Version, and Disk Space

✅ 1. Verify OS Version, Kernel Version, and Disk Space



The screenshot shows a terminal window with the following commands and output:

```
[root@localhost ~]#  
[root@localhost ~]# cat /etc/redhat-release  
Red Hat Enterprise Linux release 9.6 (Plow)  
[root@localhost ~]# uname -r  
5.14.0-570.17.1.el9_6.x86_64  
[root@localhost ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	4.0M	0	4.0M	0%	/dev
tmpfs	870M	0	870M	0%	/dev/shm
tmpfs	348M	1.5M	347M	1%	/run
/dev/nvme0n1p3	18G	7.4G	11G	42%	/
/dev/nvme0n1p1	295M	218M	78M	74%	/boot
tmpfs	174M	52K	174M	1%	/run/user/42
tmpfs	174M	36K	174M	1%	/run/user/1000

✓ 2. Execute the Patching Script

Execute the automated patching script as the root user to initiate the patching process.

```
# ./rhel9_patching.sh
```

On execution, the script initializes logging and displays system details such as date, hostname, OS version, kernel version, and log file location.

```
[root@localhost ~]# ./rhel9_patching.sh
=====
RHEL 9 Advanced Patching Started
Date       : Sun Jan 25 10:14:50 AM PST 2026
Hostname   : localhost.localdomain
OS Version : Red Hat Enterprise Linux release 9.6 (Plow)
Kernel Before Patch : 5.14.0-570.17.1.el9_6.x86_64
Log File   : /var/log/patching/patching_2026-01-25_10-14.log
=====
[WARNING] This will APPLY patches on this system. Continue? (yes/no): yes
[INFO] Checking Red Hat subscription...
=====+
System Status Details
+-----+
Overall Status: Disabled
Content Access Mode is set to Simple Content Access. This host has access to content, regardless of subscription status.

System Purpose Status: Disabled

[INFO] Checking enabled repositories...
Updating Subscription Management repositories.
repo id      repo name
rhel-9-for-x86_64-appstream-rpms Red Hat Enterprise Linux 9 for x86_64 - AppStream (RPMs)
rhel-9-for-x86_64-baseos-rpms   Red Hat Enterprise Linux 9 for x86_64 - BaseOS (RPMs)
[INFO] Capturing pre-patch installed packages...
[INFO] Checking available updates...
Updating Subscription Management repositories.
Last metadata expiration check: 0:41:05 ago on Sun 25 Jan 2026 09:33:58 AM PST.

NetworkManager.x86_64      1:1.54.0-3.el9_7      rhel-9-for-x86_64-baseos-rpms
NetworkManager-adsI.x86_64 1:1.54.0-3.el9_7      rhel-9-for-x86_64-baseos-rpms
NetworkManager-libnm.x86_64 1:1.54.0-3.el9_7      rhel-9-for-x86_64-baseos-rpms
NetworkManager-team.x86_64 1:1.54.0-3.el9_7      rhel-9-for-x86_64-baseos-rpms
NetworkManager-tui.x86_64  1:1.54.0-3.el9_7      rhel-9-for-x86_64-baseos-rpms
NetworkManager-wifi.x86_64 1:1.54.0-3.el9_7      rhel-9-for-x86_64-baseos-rpms
aardvark-dns.x86_64        2:1.16.0-1.el9        rhel-9-for-x86_64-appstream-rpms
alsa-lib.x86_64            1:2.14-1.el9          rhel-9-for-x86_64-appstream-rpms
alsa-ucm.noarch            1:2.14-1.el9          rhel-9-for-x86_64-appstream-rpms
alsa-utils.x86_64          1:2.14-2.el9          rhel-9-for-x86_64-appstream-rpms
anaconda-core.x86_64       34.25.7.6-1.el9       rhel-9-for-x86_64-appstream-rpms
anaconda-gui.x86_64        34.25.7.6-1.el9       rhel-9-for-x86_64-appstream-rpms
anaconda-tui.x86_64        34.25.7.6-1.el9       rhel-9-for-x86_64-appstream-rpms
```

✓ 3. Apply RHEL 9 Patches Using DNF (Automated)

After successful pre-check validations, the automated patching script initiates the patching process using DNF.

```
[INFO] Starting patching...
Updating Subscription Management repositories.
Last metadata expiration check: 0:41:07 ago on Sun 25 Jan 2026 09:33:58 AM PST.
Dependencies resolved.
=====
Package                               Arch    Version                               Repository                               Size
=====
Installing:
kernel                               x86_64  5.14.0-611.24.1.el9_7               rhel-9-for-x86_64-baseos-rpms          1.1 M
kernel-core                          x86_64  5.14.0-611.24.1.el9_7               rhel-9-for-x86_64-baseos-rpms          17 M
kernel-devel                         x86_64  5.14.0-611.24.1.el9_7               rhel-9-for-x86_64-appstream-rpms       22 M
kernel-modules                      x86_64  5.14.0-611.24.1.el9_7               rhel-9-for-x86_64-baseos-rpms          40 M
kernel-modules-core                  x86_64  5.14.0-611.24.1.el9_7               rhel-9-for-x86_64-baseos-rpms          31 M
Upgrading:
NetworkManager                     x86_64  1:1.54.0-3.el9_7                    rhel-9-for-x86_64-baseos-rpms          2.4 M
NetworkManager-adsl                 x86_64  1:1.54.0-3.el9_7                    rhel-9-for-x86_64-baseos-rpms           32 k
NetworkManager-libnm                x86_64  1:1.54.0-3.el9_7                    rhel-9-for-x86_64-baseos-rpms          1.9 M
NetworkManager-team                  x86_64  1:1.54.0-3.el9_7                    rhel-9-for-x86_64-baseos-rpms           37 k
NetworkManager-tui                   x86_64  1:1.54.0-3.el9_7                    rhel-9-for-x86_64-baseos-rpms          247 k
NetworkManager-wifi                  x86_64  1:1.54.0-3.el9_7                    rhel-9-for-x86_64-baseos-rpms           81 k
aardvark-dns                         x86_64  2:1.16.0-1.el9                      rhel-9-for-x86_64-appstream-rpms        874 k
alsa-lib                             x86_64  1.2.14-1.el9                        rhel-9-for-x86_64-appstream-rpms        529 k
alsa-ucm                             x86_64  1.2.14-1.el9                        rhel-9-for-x86_64-appstream-rpms        424 k
alsa-utils                           x86_64  1.2.14-2.el9                        rhel-9-for-x86_64-appstream-rpms        1.2 M
anaconda-core                        x86_64  34.25.7.6-1.el9                     rhel-9-for-x86_64-appstream-rpms        2.7 M
anaconda-gui                         x86_64  34.25.7.6-1.el9                     rhel-9-for-x86_64-appstream-rpms        416 k
anaconda-tui                         x86_64  34.25.7.6-1.el9                     rhel-9-for-x86_64-appstream-rpms        123 k
anaconda-widgets                     x86_64  34.25.7.6-1.el9                     rhel-9-for-x86_64-appstream-rpms         52 k
at                                    x86_64  3.1.23-13.el9_7                     rhel-9-for-x86_64-baseos-rpms           63 k
audit                                x86_64  3.1.5-7.el9                          rhel-9-for-x86_64-baseos-rpms          261 k
=====
```

During the patching process, DNF performs package downloads, dependency checks, and transaction validation before applying updates.

```
Transaction Summary
=====
Install 17 Packages
Upgrade 594 Packages
Remove 1 Package

Total download size: 1.5 G
Downloading Packages:
(1/611): elfutils-devel-0.193-1.el9.x86_64.rpm 50 kB/s | 50 kB 00:01
(2/611): xz-devel-5.2.5-8.el9_0.x86_64.rpm 35 kB/s | 59 kB 00:01
(3/611): spirv-tools-libs-2025.2-1.el9.x86_64.r 823 kB/s | 1.5 MB 00:01
(4/611): tbb-2020.3-9.el9.x86_64.rpm 53 kB/s | 172 kB 00:03
(5/611): llvm-filesystem-20.1.8-3.el9.x86_64.rp 23 kB/s | 9.2 kB 00:00
(6/611): systemtap-devel-5.3-3.el9.x86_64.rpm 721 kB/s | 2.4 MB 00:03
(7/611): systemtap-runtime-5.3-3.el9.x86_64.rpm 509 kB/s | 457 kB 00:00
(8/611): boost-filesystem-1.75.0-13.el9_7.x86_6 114 kB/s | 55 kB 00:00
(9/611): dyninst-13.0.0-1.el9.x86_64.rpm 985 kB/s | 3.7 MB 00:03
(10/611): systemtap-client-5.3-3.el9.x86_64.rpm 1.3 MB/s | 4.1 MB 00:03
(11/611): mokutil-0.7.2-1.el9.x86_64.rpm 127 kB/s | 47 kB 00:00
```

```
(607/611): kernel-tools-5.14.0-611.24.1.el9_7.x 930 kB/s | 1.4 MB 00:01
(608/611): kernel-tools-libs-5.14.0-611.24.1.el 969 kB/s | 1.1 MB 00:01
(609/611): glib2-2.68.4-18.el9_7.1.x86_64.rpm 944 kB/s | 2.6 MB 00:02
(610/611): iwl7260-firmware-25.30.13.0-155.1.el 1.4 MB/s | 67 MB 00:49
(611/611): linux-firmware-20251111-155.1.el9_7. 2.3 MB/s | 602 MB 04:17
-----
Total 2.5 MB/s | 1.5 GB 10:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Running scriptlet: linux-firmware-20251111-155.1.el9_7.noarch 1/1
Running scriptlet: selinux-policy-targeted-38.1.65-1.el9.noarch 1/1
Running scriptlet: java-1.8.0-openjdk-headless-1:1.8.0.472.b08-2.el9.x8 1/1
Running scriptlet: firewall-1.3.4-15.el9_6.noarch 1/1
Running scriptlet: java-17-openjdk-headless-1:17.0.18.0.8-1.el9.x86_64 1/1
Running scriptlet: kmod-kvdo-8.2.6.3-175.el9_7.x86_64 1/1
Preparing : 1/1
Upgrading : linux-firmware-updates-20251111-155.1.el9_7.noarch 1/1
```

✓ 4. Post-Patch Validation and Kernel Cleanup

After the patching process completes, the script performs post-patch validation and kernel cleanup.

```
Dependencies resolved.
Nothing to do.
Complete!
[INFO] Kernel after patch (before reboot): 5.14.0-570.17.1.el9_6.x86_64
[INFO] Retaining last 2 kernels...
Updating Subscription Management repositories.
Dependencies resolved.
=====
Package                Arch      Version                               Repository                               Size
=====
Removing:
kernel                  x86_64    4.18.0-553.52.1.el8_10 @rhel-8-for-x86_64-baseos-rpms 0
kernel-core             x86_64    4.18.0-553.52.1.el8_10 @rhel-8-for-x86_64-baseos-rpms 71 M
kernel-devel            x86_64    4.18.0-553.52.1.el8_10 @rhel-8-for-x86_64-baseos-rpms 53 M
kernel-modules          x86_64    4.18.0-553.52.1.el8_10 @rhel-8-for-x86_64-baseos-rpms 25 M

Transaction Summary
=====
Remove 4 Packages

Freed space: 149 M
Running transaction check
Waiting for process with pid 43458 to finish
```

✓ 5. Post-Patch Validation, Kernel Cleanup, and Reboot Check

After applying patches, the script performs final validation steps and determines whether a system reboot is required.

```
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Erasing   : kernel-4.18.0-553.52.1.el8_10.x86_64 1/4
Running scriptlet: kernel-4.18.0-553.52.1.el8_10.x86_64 1/4
  Erasing   : kernel-devel-4.18.0-553.52.1.el8_10.x86_64 2/4
  Erasing   : kernel-modules-4.18.0-553.52.1.el8_10.x86_64 3/4
Running scriptlet: kernel-modules-4.18.0-553.52.1.el8_10.x86_64 3/4
Running scriptlet: kernel-core-4.18.0-553.52.1.el8_10.x86_64 4/4
  Erasing   : kernel-core-4.18.0-553.52.1.el8_10.x86_64 4/4
Running scriptlet: kernel-core-4.18.0-553.52.1.el8_10.x86_64 4/4
  Verifying : kernel-4.18.0-553.52.1.el8_10.x86_64 1/4
  Verifying : kernel-core-4.18.0-553.52.1.el8_10.x86_64 2/4
  Verifying : kernel-devel-4.18.0-553.52.1.el8_10.x86_64 3/4
  Verifying : kernel-modules-4.18.0-553.52.1.el8_10.x86_64 4/4
Installed products updated.

Removed:
kernel-4.18.0-553.52.1.el8_10.x86_64
kernel-core-4.18.0-553.52.1.el8_10.x86_64
kernel-devel-4.18.0-553.52.1.el8_10.x86_64
kernel-modules-4.18.0-553.52.1.el8_10.x86_64

Complete!
[INFO] Checking if reboot is required...
[INFO] Reboot is NOT required
=====
          PATCHING SUMMARY
=====
Execution Mode : APPLY
Reboot Required : NO
Log File      : /var/log/patching/patching_2026-01-25_10-42.log
=====
[INFO] Sending patching report to awasthirish1999@gmail.com
[root@localhost ~]#
```

✓ 6. Patching Summary and Email Notification

Once patching is completed, the script generates a summary report and sends a detailed patching log via email.

```
Complete!
[INFO] Checking if reboot is required...
[INFO] Reboot is NOT required

=====
                        PATCHING SUMMARY
=====
Execution Mode   : APPLY
Reboot Required  : NO
Log File         : /var/log/patching/patching_2026-01-25_10-42.log
=====
[INFO] Sending patching report to awasthirishi999@gmail.com
[root@localhost ~]#
```

✓ 7. Verify Patching and Kernel Cleanup Processes

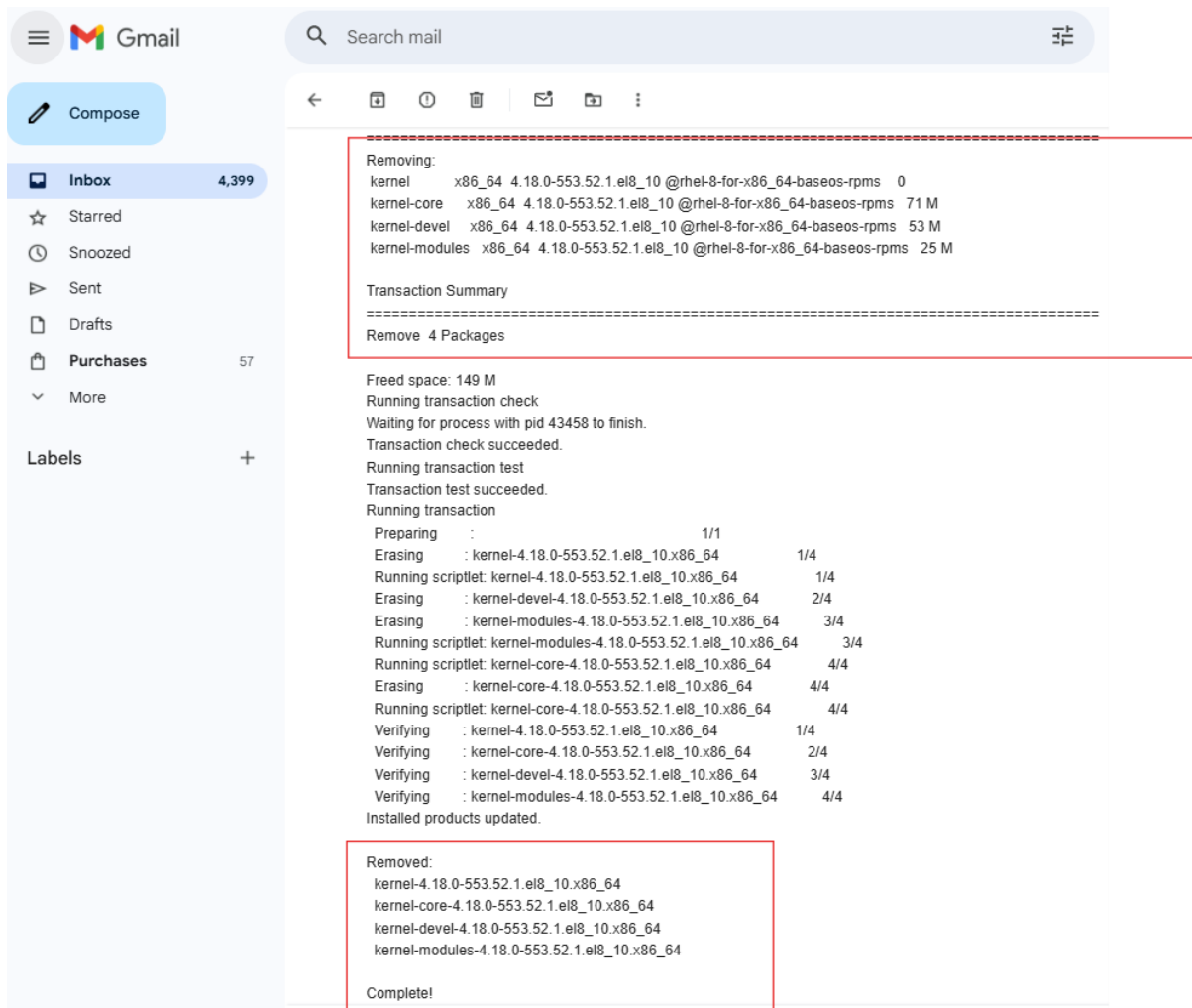
After patching and kernel cleanup, the system is checked to ensure no active or stuck DNF processes are running.

To validate this, running processes are verified using:

```
# ps aux | grep -E 'yum|dnf'
```

```
[root@localhost ~]#
[root@localhost ~]# ps aux | grep -E 'yum|dnf'
root      43458  13.7  6.7 781744 120288 ?        S   10:15   4:41 /usr/bin/python3.9 /usr/bin/dnf update -y
root      88861   0.8  7.6 377096 136060 pts/1    S+  10:43   0:03 /usr/bin/python3.9 /usr/bin/dnf remove -y kernel-0:4.18.0-553.52.1.el8_10.x86_64 kernel-core-0:4.18.0-553.52.1.el8_10
.x86_64 kernel-devel-0:4.18.0-553.52.1.el8_10.x86_64 kernel-modules-0:4.18.0-553.52.1.el8_10.x86_64
root      148837  0.0  0.1 221804 2432 pts/2      S+  10:49   0:00 grep --color=auto -E yum|dnf
[root@localhost ~]#
[root@localhost ~]#
```

✓ 8. Received mail after completion of patching



Complete!

[INFO] Checking if reboot is required...

[INFO] Reboot is NOT required

```
=====
PATCHING SUMMARY
=====
Execution Mode : APPLY
Reboot Required : NO
Log File       : /var/log/patching/patching_2026-01-25_10-42.log
=====
[INFO] Sending patching report to awasthirishi999@gmail.com
```

✅ Below is the complete automation patching script used in this project:

```
#!/bin/bash
```

```
# =====
```

```
# Advanced RHEL 9 Patching Script
```

```
# Author: Rishabh Awasthi
```

```
# Features: Logging, Reboot Check, Kernel Cleanup,
```

```
#     Email Notification (Postfix + s-nail)
```

```
# =====
```

```
#####
```

```
# CONFIGURATION
```

```
#####
```

```
LOG_DIR="/var/log/patching"
```

```
TIMESTAMP="$(date +%F_%H-%M)"
```

```
LOG_FILE="$LOG_DIR/patching_${TIMESTAMP}.log"
```

```
EMAIL="awasthirishi999@gmail.com" # change if needed
```

```
AUTO_REBOOT=false
```

```
KEEP_KERNELS=2
```

```
#####
```

```
# INTERNAL FUNCTION
```

```
#####
```

```
run_cmd() {
```

```
    eval "$@"
```

```
}
```

```
#####
```

```
# SETUP LOGGING
```

```
#####
```

```
mkdir -p "$LOG_DIR"
```

```
exec >>(tee -a "$LOG_FILE") 2>&1
```

```
echo "=====
```

```
echo "    RHEL 9 Advanced Patching Started"
```

```
echo "Date      : $(date)"
```

```
echo "Hostname   : $(hostname)"
```

```
echo "OS Version  : $(cat /etc/redhat-release)"
```

```
echo "Kernel Before Patch : $(uname -r)"
```

```
echo "Log File    : $LOG_FILE"
```

```
echo "=====
```

```
#####
```

SAFETY CONFIRMATION

```
#####
```

```
read -rp "WARNING: This will APPLY patches on this system. Continue? (yes/no): " CONFIRM
```

```
if [ "$CONFIRM" != "yes" ]; then
```

```
    echo "[INFO] Execution cancelled by user."
```

```
    exit 0
```

```
fi
```

```
#####
```

ROOT CHECK

```
#####
```

```
if [ "$EUID" -ne 0 ]; then
```

```
    echo "[ERROR] Script must be run as root"
```

```
    exit 1
```

```
fi
```

```
#####
```

CHECK REQUIRED COMMANDS

```
#####
```

```
for cmd in needs-restarting postfix; do
    if ! command -v "$cmd" &>/dev/null; then
        echo "[ERROR] Required command '$cmd' not found."
        exit 1
    fi
done
```

```
#####
```

DETECT MAIL CLIENT (s-nail / mailx / mail)

```
#####
```

```
MAIL_CMD="$(command -v mail || command -v mailx || true)"
```

```
if [ -z "$MAIL_CMD" ]; then
    echo "[WARN] No mail client found. Email notification will be skipped."
fi
```

```
#####
```

PRE-CHECKS

```
#####
```

```
echo "[INFO] Checking Red Hat subscription..."
subscription-manager status || { echo "[ERROR] System not registered with Red Hat"; exit 1; }
```

```
echo "[INFO] Checking enabled repositories..."
dnf repolist enabled || { echo "[ERROR] No enabled repositories found"; exit 1; }
```

```
#####
```

PRE-PATCH SNAPSHOT

```
#####
```

```
echo "[INFO] Capturing pre-patch installed packages..."
dnf list installed > "$LOG_DIR/installed_packages_before_${TIMESTAMP}.txt"
```

```
#####
```

UPDATE PREVIEW

```
#####
```

```
echo "[INFO] Checking available updates..."
```

```
dnf check-update || true
```

```
#####
```

PATCHING

```
#####
```

```
echo "[INFO] Starting patching..."
```

```
run_cmd "dnf update -y"
```

```
#####
```

POST-PATCH SNAPSHOT

```
#####
```

```
echo "[INFO] Kernel after patch (before reboot): $(uname -r)"
```

```
dnf list installed > "$LOG_DIR/installed_packages_after_$TIMESTAMP.txt"
```

```
#####
```

KERNEL CLEANUP

```
#####
```

```
echo "[INFO] Retaining last $KEEP_KERNELS kernels..."
```

```
dnf remove -y $(dnf repoquery --installonly --latest-limit=-$KEEP_KERNELS -q) || true
```

```
#####
```

REBOOT CHECK

```
#####
```

```
echo "[INFO] Checking if reboot is required..."
```

```
REBOOT_REQUIRED="NO"
```

```
if needs-restarting -r &>/dev/null; then
```

```

    echo "[INFO] Reboot is REQUIRED"

    REBOOT_REQUIRED="YES"
else
    echo "[INFO] Reboot is NOT required"
fi

#####

# AUTO-REBOOT (Optional)
#####

if [ "$AUTO_REBOOT" = true ] && [ "$REBOOT_REQUIRED" = "YES" ]; then
    echo "[INFO] Auto-reboot enabled. Rebooting in 1 minute..."
    shutdown -r +1 "System reboot after patching"
fi

#####

# SUMMARY
#####

echo "=====
echo "    PATCHING SUMMARY"
echo "=====
echo "Execution Mode : APPLY"
echo "Reboot Required : $REBOOT_REQUIRED"
echo "Log File      : $LOG_FILE"
echo "=====

#####

# EMAIL NOTIFICATION
#####

if [ -n "$MAIL_CMD" ] && [ -n "$EMAIL" ]; then
    SUBJECT="RHEL 9 Patching Report | $(hostname) | $(date +%F)"
    echo "[INFO] Sending patching report to $EMAIL"

```

```
"$MAIL_CMD" -s "$SUBJECT" "$EMAIL" < "$LOG_FILE"
```

```
fi
```

```
exit 0
```

✅ Final Words

This RHEL 9 patching script was built to reflect real-world enterprise patching practices focusing on safety, automation, and audit readiness. From pre-checks and logging to kernel cleanup, reboot validation, and email reporting, this project strengthened my hands-on experience with production Linux systems.

🔧 Key takeaways from this project:

- Enterprise Linux patching automation
- Pre & post patch validations
- Kernel lifecycle management
- Reboot impact analysis
- Operational logging & reporting



— Rishabh Awasthi | Application Engineer